

SDI Intelligence server: who does what

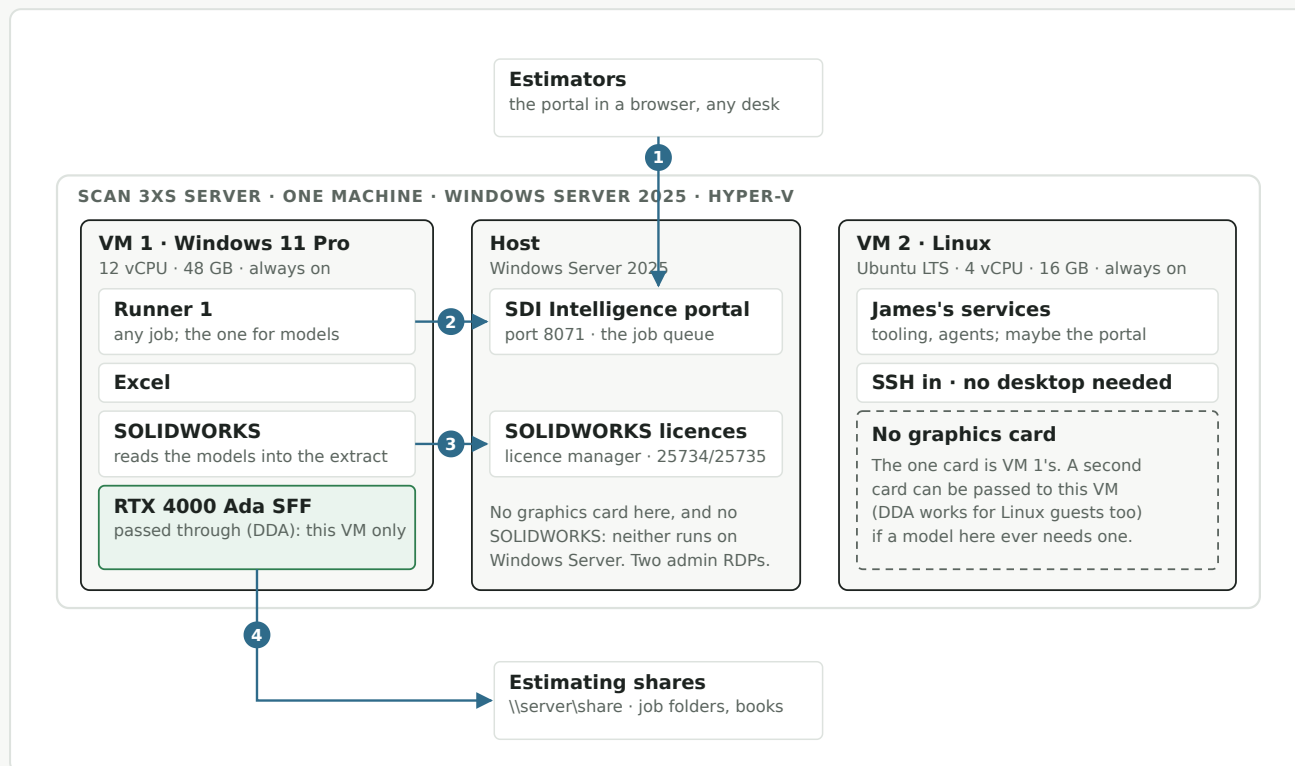
Revised 5 October 2026 · Windows Server 2025 on the machine, the graphics card and SOLIDWORKS in a Windows 11 Pro VM, and a Linux VM beside it. A Windows 11 Pro host was considered and set aside: it cannot give a graphics card to a VM.

Jack builds the machine and hands over two virtual machines that behave like new computers on the network: a Windows 11 Pro one with the graphics card inside it, and a Linux one. James takes them from there, exactly as he would set up a laptop. The handover checks in the middle are the line between the two. The first section shows how the parts fit together, why the host is Windows Server, and what to watch for.

Scan 3XS · Threadripper PRO 9975WX (32 cores) · 128 GB ECC · NVIDIA RTX 4000 Ada SFF 20 GB · 2 × 2 TB NVMe · Windows Server 2025 host · VM 1 Windows 11 Pro: the card, SOLIDWORKS, Excel, runner 1 · VM 2 Linux: James's services · optional VM 3 Windows 11 Pro for a second runner

How it fits together

One machine, split into a host and two virtual machines. The graphics card and SOLIDWORKS live in the Windows VM; the Linux VM runs the services James names; the host runs the portal and the job queue and nothing that needs a desktop.



1. Estimators start a job on the portal. It joins the queue on the host.

2. The runner in VM 1, when it is free, takes the oldest job waiting. One job at a time. The Linux VM runs no estimates: it has no Excel and no SOLIDWORKS. A second Windows VM with its own runner can be added later (VM 3).
3. SOLIDWORKS takes a licence from the pool on the host while it is open. The runner keeps it open, so VM 1 holds one seat all day.
4. The runner reads and writes the job folders and books on the estimating shares, which stay on the file server.

WHY THE HOST IS WINDOWS SERVER AND NOT WINDOWS 11 PRO

Three things were asked of this machine: Remote Desktop into it, the graphics card in a VM, and SOLIDWORKS in a VM. Only a Server host gives all three.

ASKED FOR	WINDOWS SERVER 2025 HOST	WINDOWS 11 PRO HOST
The graphics card used by a VM	Yes. Discrete Device Assignment hands the whole card to one VM, with NVIDIA's normal driver inside it.	No. DDA is a Server feature. The card can only drive the host's own desktop. The partitioning Windows 11 has (GPU-P) is unsupported by NVIDIA and Dassault for this card and does not reach Linux guests at all.
SOLIDWORKS in a VM	Yes, in VM 1 with the card. Dassault lists Hyper-V 2025 for SOLIDWORKS 2026.	Only in software graphics, which the engine can test but nobody would design on.
Remote Desktop	Two administrator sessions on the host at once. The Windows VM still allows one desktop, like any Windows 11; the Hyper-V console reaches it without logging the runner off.	One desktop on the host, which is also the runner's. Any Remote Desktop as another user pushes the runner off.
Can the card be shared between the metal and a VM?	No, on either. A card belongs to one operating system at a time. Passed through, it is VM 1's and the host has none, which is fine: a Server host needs no graphics. Sharing one card between several VMs needs a data-centre card (NVIDIA L4 and similar) with yearly vGPU licences.	
Cost	Server 2025 Standard is licensed per core, 16-core minimum: two packs for 32 cores, plus client access licences. Plus Windows 11 Pro and Office for VM 1.	Windows 11 Pro comes with the workstation.

WHAT CAN AND CANNOT BE SHARED

The graphics card: one card, one VM.

Passed through (DDA), the RTX 4000 Ada belongs to VM 1 alone; the host and the Linux VM have none. It cannot be split between VMs: Windows Server 2025 only partitions a short list of data-centre cards (NVIDIA L4, L40S, A16 and similar) with NVIDIA's licensed vGPU driver, and this card is not one of them. Moving it to another VM needs both VMs shut down, so it is a maintenance job, not something that happens per request. The host does not need it. If the Linux VM ever needs a card of its own, a second card is fitted and passed to it the same way; DDA works for Linux guests.

The estimates: shared by the queue.

This is where "whoever is free takes the next one" happens. A second Windows VM with a runner means two estimates at once; each takes the oldest job waiting when it finishes the last.

SOLIDWORKS: the licences are shared, the program is not.

Each VM that uses SOLIDWORKS needs its own installation; it cannot run on the host and be reached from the VMs, and the engine drives it only on its own VM. A network (SolidNetWork) licence puts the seats in a pool on the host: SOLIDWORKS takes one when it opens and returns it when it closes, first come, first served, and designers' PCs can share the same pool. A standalone licence works in a VM but is tied to that VM's identity, so a rebuilt VM means a reactivation; the network licence avoids that.

The Linux VM: James's, with no desktop.

Ubuntu LTS server under Hyper-V, administered over SSH. It runs tooling, agents and any service that needs no Windows; the portal could move to it (it is a Python service and runs the same on Linux). Nothing that drives Excel or SOLIDWORKS goes here. Ubuntu carries the Hyper-V integration drivers in its kernel, so nothing extra is installed.

Test before buying another card.

The model extract may not need the graphics card at all: its geometry work runs on the processor, and SOLIDWORKS falls back to software graphics without a card. Run the extractor on job 12120 in a VM with no card and compare the result and the time with VM 1. If they match, a second Windows VM can run SOLIDWORKS without a card, and no GPU purchase is needed.

IF MORE GRAPHICS CAPACITY IS EVER NEEDED

OPTION	WHAT IT GIVES	CATCH
A second RTX 4000 Ada SFF, passed through to the Linux VM or to VM 3	A graphics card in a second VM	Supported and simple. Scan to confirm a free slot and power.
Replace with a card that partitions (NVIDIA L4)	Three or more VMs, each with a fixed slice	An NVIDIA vGPU licence per VM each year; needs server-style cooling; the host still gets no share; SOLIDWORKS support to confirm.

OPTION	WHAT IT GIVES	CATCH
Share this card through an unsupported driver trick	Host and VMs share one card	Needs a Windows 11 host; unsupported by Microsoft, NVIDIA and Dassault; breaks on driver updates. Not for this machine.

Gotchas

None of these stops the plan. Each one is a setting or a habit, and each is on a list below with an owner.

1. **The card is VM 1's, and touching it means stopping VM 1.** A driver update inside the VM is an ordinary restart of the VM. Moving the card, or changing which VM has it, needs the VMs shut down. The host shows "Microsoft Basic Display" from then on; that is expected.
2. **One desktop in the Windows VM.** VM 1 is Windows 11 and allows one interactive session. A Remote Desktop into it as a different user pushes the runner off and can stall an estimate. Rule: connect as the runner account (same session) and leave by handing the session back to the console with `tscon`, or use the Hyper-V console from the host, which shows the runner's desktop without logging it off. The host's own two admin sessions are the normal way in.
3. **Server drivers for a workstation board.** Scan supplies Windows 11 drivers for the 3XS. Windows Server 2025 installs on the Threadripper PRO board, but chipset and network drivers may need to come from the board maker. Ask Scan to confirm before the build week.
4. **SOLIDWORKS version and licence.** 2026 or later for Hyper-V 2025 support. A network licence is strongly preferred in a VM: a standalone one is tied to the VM's identity and reactivates on a rebuild.
5. **The VMs must be on the office network, not behind the host.** Create an External virtual switch on the machine's network port so both VMs get fixed addresses and DNS names, and the portal, Remote Desktop and SSH reach them by name.
6. **Sleep and lock in VM 1.** Windows 11 defaults: no sleep, no hibernate, no screen lock, no screensaver for the runner account, and an exemption from any domain lock policy. Auto-logon of a domain account needs Sysinternals Autologon with Windows Hello off for that account.
7. **Linux under Hyper-V.** A Generation 2 Linux VM boots with Secure Boot only on the "Microsoft UEFI Certificate Authority" template; the Windows template refuses it. No desktop and no Enhanced Session: SSH.
8. **A host restart takes every VM down.** Windows Update on the host is calmer than Windows 11 but a restart mid-run still loses that estimate. Host and VMs on different nights, and James told first.
9. **Licences add up.** Server 2025 Standard per core (two 16-core packs for this processor) and client access licences; Windows 11 Pro and Office for VM 1 (and VM 3 if built); nothing for the Linux VM. Windows Server Standard's two free guest licences are for Windows *Server* guests, not Windows 11.
10. **Keep the estimating shares on the file server.** Nothing here needs this machine to be a file server, and making it one puts the queue and the files on one restart.

JACK

Infrastructure: the host and the VMs

Everything below the applications: hardware, operating systems, network, accounts, backup. In build order.

THE HOST (WINDOWS SERVER 2025)

1. **BIOS:** latest firmware; power on after AC loss; enable virtualisation (SVM) and IOMMU, which the GPU pass-through needs.
2. **Storage:** set up the two NVMe drives (mirror or split OS / VMs — Jack's call) and install Windows Server 2025 with current drivers, from the board maker where Scan has none for Server.
3. **Network and domain:** fixed IP, DNS name, join the domain.
4. **Hyper-V:** add the Hyper-V role and create an **External virtual switch** on the machine's network port, so the VMs are on the office network.
5. **Service account** for the SDI Intelligence portal service, with read/write to the estimating shares by UNC path (`\\server\share` , not K:).
6. **Firewall:** allow the portal port (`8071`) from the office network and from the VMs.
7. **SOLIDWORKS licence manager** (SolidNetWork): install it on the host with the reseller's serial, and open ports `25734` and `25735` to the VMs (and to designers' PCs if they share the pool). Windows Server is supported for the licence manager; it needs SOLIDWORKS 2025 SP3 or later on Server 2025, and must be at least as new as the SOLIDWORKS installs.

VM 1 (WINDOWS 11 PRO): THE GRAPHICS CARD, SOLIDWORKS AND THE RUNNER

8. **Create it as Generation 2** with Secure Boot and a virtual TPM (Windows 11 requires both).
9. **Fixed resources:** 12 virtual CPUs, 48 GB memory with Dynamic Memory **off**, a 300 GB+ virtual disk on NVMe.
10. **Pass the RTX 4000 through to the VM** (Discrete Device Assignment), then install NVIDIA's RTX workstation driver inside the VM. GPU partitioning is not needed (and not available for this card). The host and the Linux VM then have no graphics card; that is expected.
11. **Windows 11 Pro**, fixed IP and DNS name, joined to the domain, fully updated.
12. **Runner account:** a dedicated domain user that logs on automatically at boot (Sysinternals Autologon, Windows Hello off for it); read/write to the estimating shares by UNC path.
13. **Power and lock policy:** no sleep, no screen lock, no screensaver on the VM, exempt from any domain lock policy — SOLIDWORKS and Excel need a live desktop session.
14. **Admin access for James:** Remote Desktop to the host (two admin sessions) and the Hyper-V console to the VM; the host's admin share (the laptop's `push-to-server.ps1` copies through it); a `tscon` hand-back script on the runner's desktop.

VM 2 (LINUX): JAMES'S SERVICES

15. **Create it as Generation 2** with Secure Boot on the **Microsoft UEFI Certificate Authority** template; no virtual TPM needed. 4 virtual CPUs, 16 GB memory, a 100 GB+ virtual disk. With VM 1 at 48 GB this leaves the host about 64 GB.
16. **Ubuntu LTS server**, fixed IP and DNS name on the External switch, SSH open to James, an account for him with sudo. **No graphics card** unless a second card is fitted and passed to it.

17. **Both VMs start automatically** with the host, VM 1 first.

VM 3 (WINDOWS 11 PRO): A SECOND RUNNER, OPTIONAL AND LATER

18. **Same as VM 1** (Generation 2, Secure Boot, virtual TPM, Windows 11 Pro, fixed IP, domain, its own runner account with auto-logon, no sleep or lock), but **no graphics card and no SOLIDWORKS**; 8 virtual CPUs, 32 GB with Dynamic Memory off, 200 GB+ disk.
19. **Only after James's routing change** (see his list), so jobs with SOLIDWORKS models always go to VM 1.

RUNNING IT

20. **Licences:** Windows Server 2025 Standard per core (two 16-core packs for the 9975WX) and client access licences; a Windows 11 Pro licence and an Office licence for **each** Windows VM; nothing for the Linux VM. Server Standard's two guest licences cover Windows *Server* VMs only.
21. **Backups:** host and VMs (Windows Server Backup or the company tool, with the VMs exported), one restore tested before go-live.
22. **Windows Update:** restarts scheduled out of hours, host and VMs on different nights, James told first.
23. **Antivirus:** apply the exclusions James supplies (SOLIDWORKS temp and the output folders).
24. **Power:** on the UPS, and set to start again after a power cut, with the VMs starting automatically.

Handover checks

Jack confirms these, then the machines are James's:

- In VM 1's Device Manager the graphics card shows as **NVIDIA RTX 4000 Ada** with the NVIDIA driver, not "Microsoft Basic Display".
- After a full restart of the host, both VMs start by themselves; VM 1 logs in as the runner account with nobody touching it and is still unlocked an hour later.
- Signed in as the runner account, VM 1 can open and save a file in the estimating folders by UNC path.
- From VM 1 and from a desk, the host answers on port 8071 (once James's service is running, this is just a browser test).
- From VM 1, the licence manager on the host answers on port 25734.
- From James's laptop: Remote Desktop to the host as an administrator while another admin is already connected; the Hyper-V console shows VM 1's desktop; SSH into the Linux VM by name.

JAMES

Applications: the laptop premise

Everything that makes it an SDI Intelligence machine, installed the same way as on the laptop.

ON VM 1 (RUNS THE ESTIMATES)

1. **SOLIDWORKS:** install on VM 1 only and point it at the host's licence manager. Open one model to confirm the graphics card is in use.
2. **Office / Excel:** install and activate; open one estimate book.
3. **PDM / Design Vault client,** if models come from the vault.
4. **Python and Git,** clone the repository, create the engine's `.venv`.
5. **The `.env` file,** written by hand for this machine — never copied from the laptop.
6. **The runner:** scheduled task at logon (`restart-runner.ps1`) pointing at the host's service.
7. **Hand Jack the antivirus exclusion paths.**

ON THE HOST (RUNS THE PORTAL)

8. **Python** and the service's own `.venv`, the backend files, and the host's own `.env`.
9. **Install and start the SDIIntelligence service;** check the portal loads from a desk.

ON THE LINUX VM

10. **What it runs is James's to name:** tooling, agents, and any service that needs no Windows. If the portal moves here, the runners point at the Linux VM's name and it runs under systemd; it needs no access to the estimating shares, which only the runners read and write.

PROVING IT

11. **Run one known job** (12645 or 12173) on VM 1 and compare the book and the run time with the laptop's.
12. **Restart the host** and confirm the portal, both VMs and the runner all come back without anyone logging in.
13. **Remote in and out once** through the Hyper-V console while a short job runs on VM 1, and confirm the job finished.
14. **The graphics test:** run the SOLIDWORKS extractor on 12120 in a VM with no card and compare the extract and the time with VM 1. This decides whether a second card is ever needed.

BEFORE VM 3

15. **Routing change in the engine:** today whichever runner is free takes the next job, so a job with SOLIDWORKS models could land on VM 3 and be costed from the PDFs alone. Each runner will say whether it has SOLIDWORKS, and jobs with models will wait for one that does.

Agree together before the build week

Rule of thumb: if it would be the same on any Windows machine, it's Jack's. If it only exists because this is SDI Intelligence, it's James's.

QUESTION	WHO LEADS	WHY IT MATTERS
SOLIDWORKS licence type: network (SolidNetWork) recommended, the cost of converting, and cover for unattended automated use	James, with the reseller	A network licence is the only way to share SOLIDWORKS: seats in a pool on the host, first come, first served, shared with designers if wanted. A standalone licence is tied to one VM and can need reactivating when the VM changes. Each SOLIDWORKS kept open (as the runner does) holds a seat all day.
SOLIDWORKS version: 2026 or later	James, with the reseller	SOLIDWORKS lists Hyper-V 2022 for version 2025 and Hyper-V 2025 (this host) for version 2026. The program itself runs only on Windows 10/11, never on the Server host.
Server 2025 drivers for the 3XS board	Jack, with Scan	Scan supplies Windows 11 drivers; the Server install may need the board maker's chipset and network drivers. Confirm before the build week.
What the Linux VM is for, and whether the portal moves to it	James	Decides its size, what Jack opens on the firewall, and which machine name the runners point at. Nothing about SOLIDWORKS or Excel changes either way.
A second runner VM (VM 3): now, later, or not at all	James	Two runners means two estimates at once. It needs its own Windows 11 and Office licences, and the routing change first.
A second graphics card, for the Linux VM or VM 3	James, after the graphics test; Jack and Scan on fit	Only if a second VM needs a card: SOLIDWORKS in VM 3, or a model on the Linux VM. A second RTX 4000 Ada SFF passed through is the simple option; a partitioning card (NVIDIA L4) adds yearly vGPU licences.
Machine names and IP addresses for host and VMs	Jack	The runner, the portal links and the service settings all name them.

QUESTION	WHO LEADS	WHY IT MATTERS
Who is told before Windows updates or restarts	Jack	A restart mid-run loses that estimate.